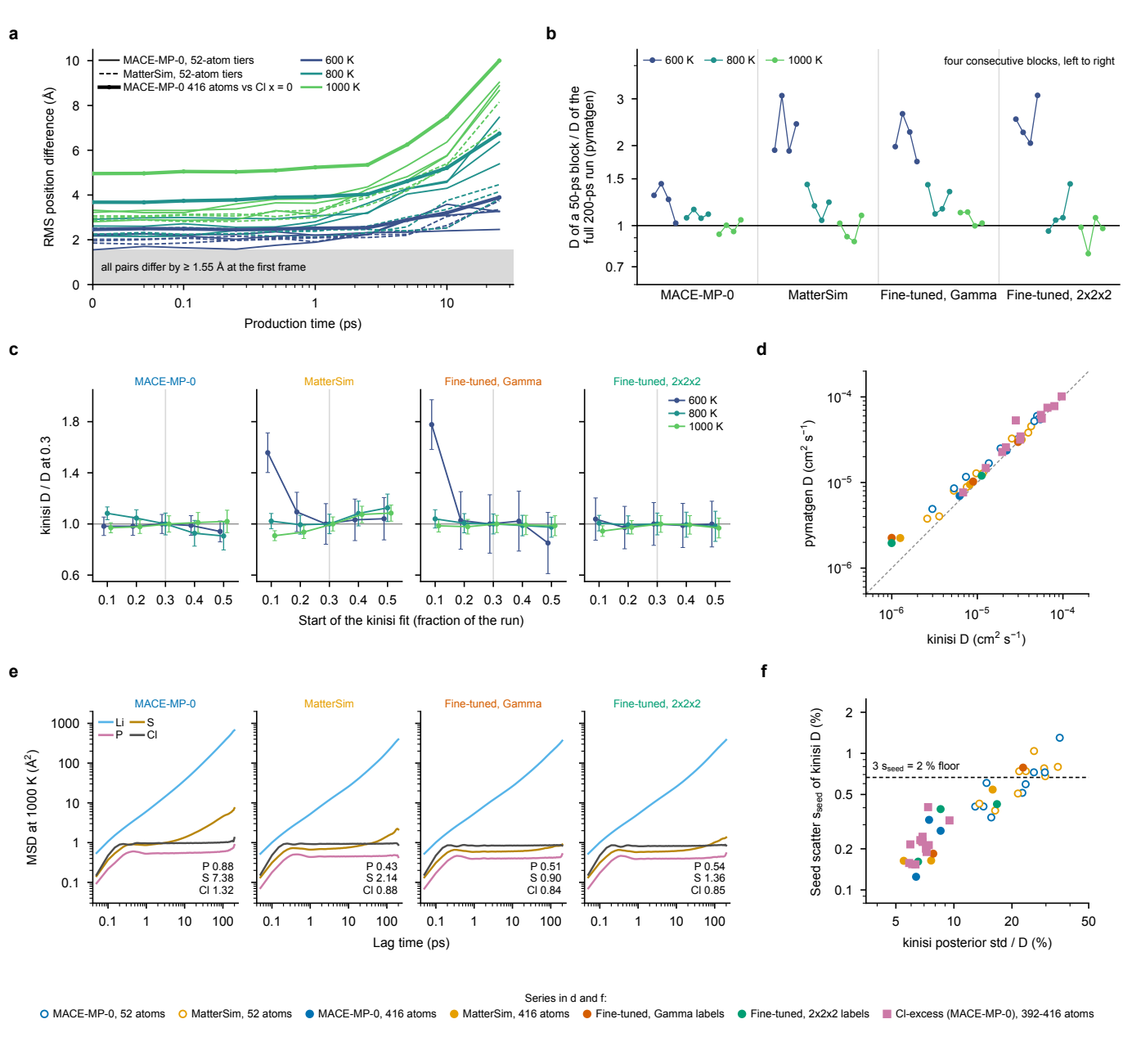

a
MACE-MP-0, 52-atom tiers
MatterSim, 52-atom tiers
MACE-MP-0 416 atoms vs Cl x = 0
600 K
800 K
1000 K
RMS position difference (Å)
all pairs differ by ≥ 1.55 Å at the first frame
Production time (ps)
b
600 K
800 K
1000 K
four consecutive blocks, left to right
D of a 50-ps block / D of the full 200-ps run (pymatgen)
MACE-MP-0
MatterSim
Fine-tuned, Gamma
Fine-tuned, 2x2x2
c
MACE-MP-0
MatterSim
Fine-tuned, Gamma
Fine-tuned, 2x2x2
600 K
800 K
1000 K
kinisi D / D at 0.3
Start of the kinisi fit (fraction of the run)
d
pymatgen D (cm² s⁻¹)
kinisi D (cm² s⁻¹)
e
MACE-MP-0
MatterSim
Fine-tuned, Gamma
Fine-tuned, 2x2x2
Li
S
P
Cl
MSD at 1000 K (Å²)
P 0.88
S 7.38
Cl 1.32
P 0.43
S 2.14
Cl 0.88
P 0.51
S 0.90
Cl 0.84
P 0.54
S 1.36
Cl 0.85
Lag time (ps)
f
3 sseed = 2 % floor
Seed scatter sseed of kinisi D (%)
kinisi posterior std / D (%)
Series in d and f:
MACE-MP-0, 52 atoms
MatterSim, 52 atoms
MACE-MP-0, 416 atoms
MatterSim, 416 atoms
Fine-tuned, Gamma labels
Fine-tuned, 2x2x2 labels
Cl-excess (MACE-MP-0), 392-416 atoms